

Spine Large-Graph Campaign Results

Correctness-Gated Spine vs. GraSU+ReGraph K1/K4-shared

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Live campaign snapshot, July 2026

Abstract

This report is generated directly from correctness-gated campaign artifacts. The snapshot is **PARTIAL**: 35 of 75 planned physical executions are currently admitted, with 4 complete three-system groups. Missing bars are unexecuted comparisons, not zero-valued measurements. All plotted rows use insertion batch 8 and exact cross-system final-state agreement.

1 End-to-end performance

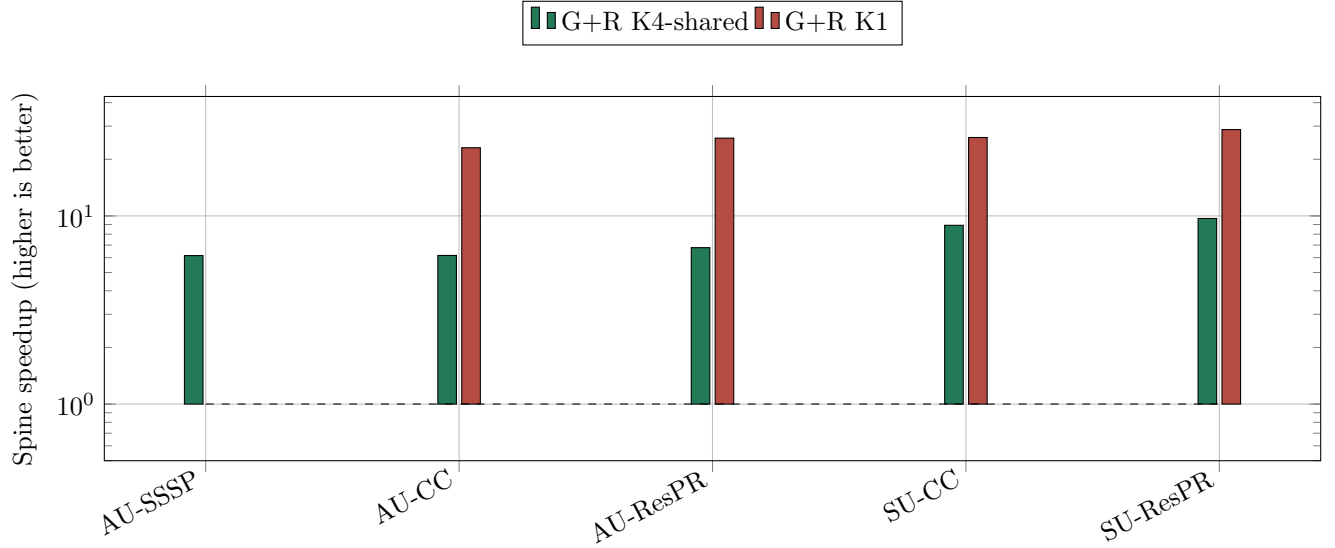


Figure 1: Device-cycle speedup after both architecture and independent mathematical correctness gates. K4-shared is the primary GraSU baseline.

2 Memory traffic and HBM energy

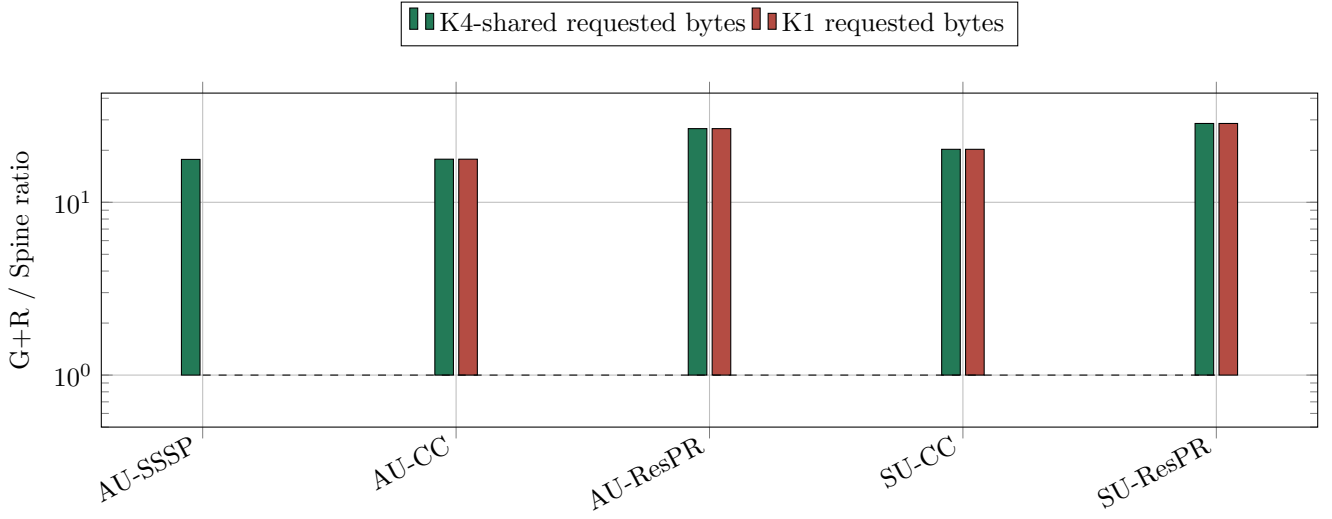


Figure 2: Accepted backend bytes. Ratios above one mean GraSU+ReGraph moves more data than Spine; request and DRAM completion ledgers close on every row.

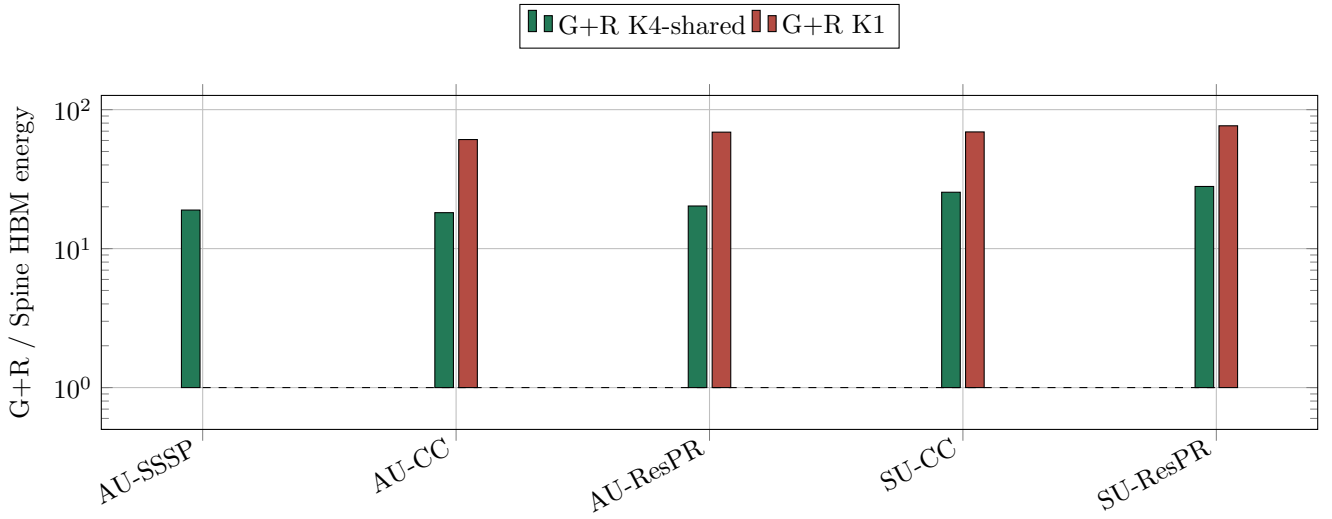


Figure 3: All-controller DRAMSim3 energy ratio. This is HBM energy, not total accelerator or board energy.

3 Update-only tradeoff

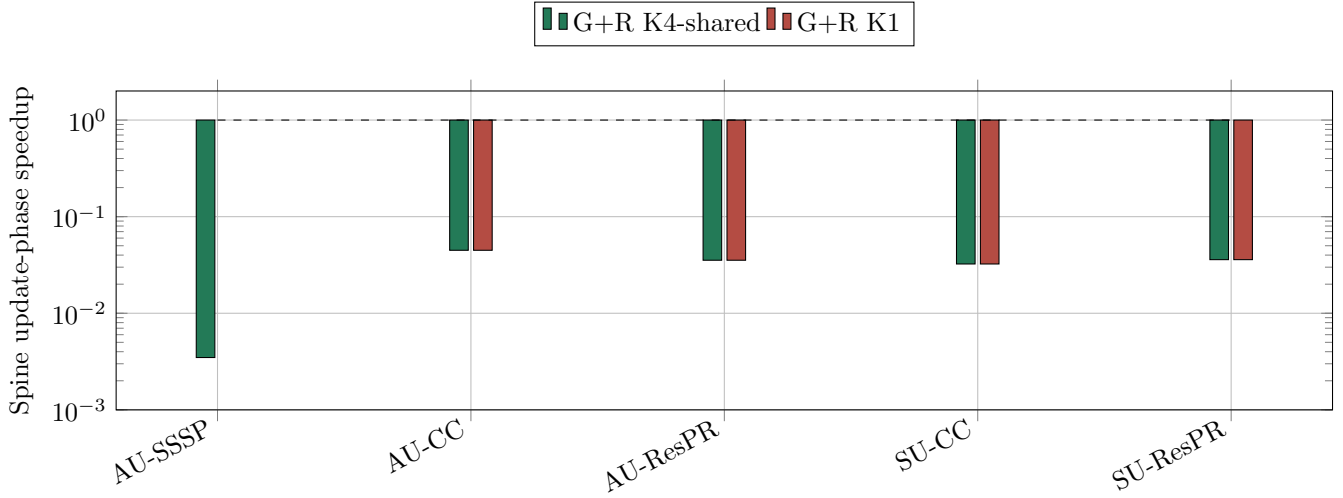


Figure 4: Update-phase cycles only. Values below one expose Spine’s current maintenance cost even when end-to-end execution favors Spine.

4 Simulator engineering runtime

System	Completed runs	Median host s	Maximum host s
Spine	13	174.4	5966.0
G+R K4-shared	6	835.5	6142.3
G+R K1	4	2087.7	2452.3

Table 1: Observed host wall time under concurrent campaign load. These values measure simulator throughput and are not accelerator performance.

Claim boundary. Only complete correctness-gated pairs enter the figures. Current FPGA/RTL evidence supports feasibility and PPA separately; this report does not claim cycle-for-cycle FPGA calibration or ASIC total power.